Table 2 (reproduction, pilot, $B = 48$ per cell). Relative bias of exposure effect estimates of cumulative total exposure with model misspecification (bold cells with correct model specification).

	Relative bias (%)					
	Naive		RRC corrected			
$\rho(c, C)$	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M
A. Mean relative bias over converged fits; the paper's values in (gray)						
Cox+Additive Cox true $\beta = 1$, additive independent error						
0.3	−29.6 (−33.5)	– (−28.3)	**394.3 (−4.4)**	201.7 (−5.7)	– (5.9)	– (6.6)
0.6	−6.6 (−32.8)	– (−27.9)	**146.4 (−4.0)**	135.7 (−5.5)	– (5.6)	– (6.1)
0.9	−0.2 (−31.4)	– (−27.2)	**22.8 (−3.8)**	43.4 (−5.4)	– (5.3)	– (5.7)
Cox+Multiplicative Cox true $\beta = 1$, multiplicative independent error						
0.3	−94.2 (−34.7)	272.5 (−29.0)	390.6 (−6.1)	**414.0 (−5.7)**	– (4.8)	– (4.3)
0.6	−30.0 (−33.9)	– (−28.4)	179.3 (−5.6)	**184.1 (−5.3)**	– (4.5)	– (4.1)
0.9	−3.6 (−32.6)	– (−27.8)	26.5 (−5.4)	**30.4 (−5.0)**	– (4.0)	– (3.7)
ERR+Additive ERR true $\beta = 1.109$, additive independent error						
0.3	−79.4 (−42.1)	−63.1 (−35.8)	70.8 (−14.9)	−13.6 (−15.2)	**213.4 (−5.5)**	487.6 (−6.6)
0.6	−70.9 (−40.6)	−33.3 (−33.0)	−23.2 (−14.5)	−34.3 (−14.9)	**858.5 (−5.1)**	717.6 (−6.4)
0.9	−71.2 (−38.9)	−22.0 (−31.2)	−64.6 (−14.2)	−60.1 (−14.7)	**193.1 (−4.6)**	446.9 (−6.2)
ERR+Multiplicative ERR true $\beta = 1.109$, multiplicative independent error						
0.3	−97.9 (−43.0)	−94.9 (−36.2)	72.4 (−16.8)	64.8 (−16.0)	236.8 (−6.7)	**338.5 (−6.3)**
0.6	−80.9 (−41.7)	26.7 (−33.4)	−22.7 (−15.8)	−26.9 (−15.6)	– (−5.8)	**– (−6.1)**
0.9	−76.7 (−40.4)	34.9 (−31.9)	−69.6 (−14.7)	−69.3 (−15.2)	176.7 (−4.8)	**181.2 (−5.7)**
B. Median relative bias over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]						
Cox+Additive Cox true $\beta = 1$, additive independent error						
0.3	−29.5	−29.5 [48]	**404.4**	207.0	4408.6 [48]	4408.6 [48]
0.6	−6.3	9.2 [48]	**144.8**	132.7	4408.6 [48]	4408.6 [48]
0.9	0.2	38.4 [48]	**22.3**	40.7	1473.7 [48]	4408.6 [48]
Cox+Multiplicative Cox true $\beta = 1$, multiplicative independent error						
0.3	−93.9	4408.6 [47]	405.3	**411.2**	4408.6 [48]	4408.6 [48]
0.6	−31.1	4408.6 [48]	173.7	**181.5**	4408.6 [48]	4408.6 [48]
0.9	−3.0	4408.6 [48]	26.6	**30.5**	4408.6 [48]	4408.6 [48]
ERR+Additive ERR true $\beta = 1.109$, additive independent error						
0.3	−78.9	−58.0 [8]	67.1	−13.3	**4408.6 [46]**	4408.6 [38]
0.6	−71.8	−12.4 [18]	−25.9	−35.5	**4408.6 [46]**	4408.6 [41]
0.9	−72.4	−15.6 [17]	−66.4	−62.0	**123.4 [17]**	1493.9 [20]
ERR+Multiplicative ERR true $\beta = 1.109$, multiplicative independent error						
0.3	−97.7	−94.9	72.9	68.8	4408.6 [44]	**4408.6 [45]**
0.6	−81.0	−19.3 [2]	−23.2	−26.5	4408.6 [48]	**4408.6 [48]**
0.9	−77.4	−14.3 [3]	−70.7	−70.7	180.4 [19]	**114.6 [16]**

Cox model $\lambda(t) = \lambda_0(t)\exp(\beta x(t))$ with true $\beta_{\text{Cox}} = 1$; ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{\text{ERR}} = 1.109$. Relative bias of each fit is against the true β of the model it fits: $100(\hat{\beta} - \beta)/\beta$.
$n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. True annual exposure c has compound symmetry ($\rho = 0.5$) across years; $E(c) = 0.05$.
– : no fit converged. A median of 4408.6 means at least half of the fits stopped at the cap $\beta = 50$ (ERR fits are not identified under the current exposure level).